Fundamentals of Deep Learning and Programming

Lecture 0: Course Overview

Fall 2025 Semester

Adjunct Professor: Donghoon Kang

Course Overview

Lecture Time	Fridays, 2 PM – 5 PM
Classroom	Room: L3 301, 302 (3334A, B)
Evaluation Criteria	Attendance (10%) Programming Assignments (15%) Term-project (15%) Mid-term (30%), Final (30%)

- Programming assignments: students should submit the code and a 2-3 page document briefly explaining the code to me by email (chocopie@kist.re.kr)

Course Website

`https://kimbabmoowoo.github.io/lecture.html`

Term Project

Must be performed alone

- Should select a paper published only in CVPR, ICCV, ECCV, or NeurIPS after 2023 (inclusive).
- Should write an 8-10 page report analyzing the training code of the selected paper.
  (including the experimental results using a subset of training data)

Before Week 7: Should decide which paper to choose. (also possible to change another paper later)

Before Week 15: Should submit an 8-10 page report as a term-project.

Schedule

September

Week 1	Course overview Introduction to Python Programming
Week 2	Useful Python Libraries — Numpy, PyTorch
Week 3	Neural Networks Basics
Week 4	Convolutional Neural Networks (CNNs)

Schedule

October

Week 5	Models using CNN	
Week 6	Introduction to Natural Language Processing (NLP)	
Week 7	Attention Mechanism	
Week 8	Transformer	Part 1
Week 9	Mid term exam	

Schedule

November

Week 10	Transformer	Part 2	
Week 11	Transformer	Part 3	Vision Transformer (ViT)
Week 12	Generative Models – Introduction to GANs, VAEs		
Week 13	Diffusion Models	Part 1	

Schedule

December

Week 14	Diffusion Models	Part 2
Week 15	Diffusion Models	Part 3
Week 16	Final exam	